LEARNER GUIDE

Naming the sides

Introduction to Trigonometry. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

In the last lesson you saw that a fixed angle in a right-angled triangle keeps the ratio of two sides constant. To use that idea, mathematicians give the three sides names — but the clever part is that two of the names depend on which angle you are looking from. In this lesson you meet the hypotenuse (the longest side, always opposite the right angle), the opposite (the side across from the angle θ) and the adjacent (the side next to θ). You label triangles in every orientation, switch the marked angle and watch the opposite and adjacent swap while the hypotenuse stays fixed, and judge other people's labels for the classic slips. The question you are exploring is: which side is which — and why does it depend on the angle you choose?

What you are learning to notice

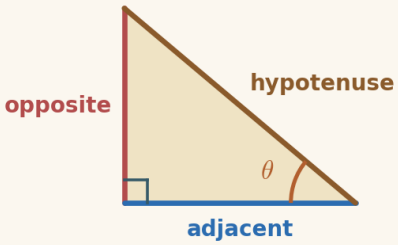
- That the hypotenuse is the longest side and is always opposite the right angle — it never changes name.
- That the opposite side is across from the marked angle θ (it does not touch θ).
- That the adjacent side is next to θ and is not the hypotenuse.
- That switching to the other acute angle makes the opposite and adjacent swap, while the hypotenuse stays the same.

Moving through it, one step at a time

In Side Labeller, watch a triangle with its three sides named from the marked angle θ; press Switch the angle to move θ to the other corner and New triangle to change orientation — notice what changes and what does not. In Label it yourself, one side is highlighted and you choose whether it is the hypotenuse, opposite or adjacent, in triangles rotated every which way. In Which side is it?, you judge a claim about a highlighted side as right or a slip — watching especially for the adjacent being mistaken for the hypotenuse. Finally, in Same triangle, two viewpoints, you name the sides of one triangle from angle A and then from angle B, and see the opposite and adjacent exchange. Use the Scratchpad to draw a triangle and label it from both angles yourself.

Worked examples

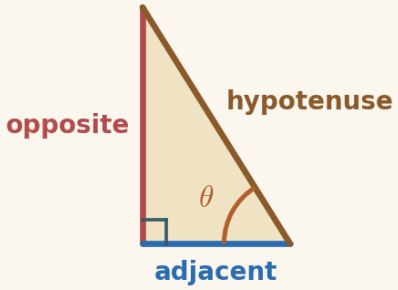
Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.



Example 1 — naming the three sides

Every right-angled triangle has three sides. Two of the names depend on where the marked angle θ is; one never moves.

1. the side opposite the right angle	the hypotenuse
2. the side across from θ	the opposite
3. the side beside θ (not the hypotenuse)	the adjacent



Example 2 — the same three questions

The shape has changed, but the naming works exactly the same way. If the OTHER acute angle were the marked one, the opposite and adjacent would swap over — the hypotenuse never does.

1. the side opposite the right angle	the hypotenuse
2. the side across from θ	the opposite
3. the side beside θ	the adjacent

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (right-angled triangles, the right angle (90°) and its square marker, the hypotenuse as the longest side, and reading a marked angle θ (from Lesson 1 and the Pythagoras unit)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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